

Primitive Cofactor Coordinates for Two-Key Erasure Rectangles: Divisor Compression, Frozen Prime Rays, and the Residual Möbius Plane

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Abstract

The first signed mixed term in a fixed-shift missing-native Gram is a two-key rectangle: one key joins two source rows through equal residual labels, while a second key joins the same rows through unequal residual labels. We give an exact arithmetic normal form for every such rectangle. For distinct primitive rows m, n , put

$$H = h_0(n - m), \quad \Delta = j_E - j_R,$$

where j_R and j_E are the same-residual and erasure times. The four targets satisfy

$$nA_m - mA_n = nB_m - mB_n = H, \quad A_mB_n - B_mA_n = H\Delta.$$

Write the fixed row divisors as

$$d_m = ga, \quad d_n = gb, \quad (a, b) = 1.$$

If the anchor cofactors obey $d_m R_m = d_n R_n$, then squarefreeness forces

$$R_m = bt, \quad R_n = at, \quad t \mid H.$$

For fixed t , all physical anchor solutions lie on one affine ray

$$j_R = j_{R,0} + abt z, \quad P_m = P_{m,0} + ma z, \quad P_n = P_{n,0} + nb z.$$

At the erasure key, write

$$q = (S_m, S_n), \quad S_m = qu, \quad S_n = qv, \quad (u, v) = 1.$$

Then $q \mid H$, and unequal residual labels are equivalent to $(u, v) \neq (b, a)$. For fixed q, u, v , all physical erasure solutions lie on one ray

$$j_E = j_{E,0} + quv z, \quad Q_m = Q_{m,0} + mv z, \quad Q_n = Q_{n,0} + nu z.$$

The four target Möbius signs reduce exactly to

$$\mu(a)\mu(b)\mu(u)\mu(v).$$

In particular, the arithmetic sign is constant along each prime ray. After retaining every native mask, coefficient, and phase in a literal kernel $K_{a,q}(u, v)$, the rectangle sum becomes the exact primitive-plane transform

$$\mu(a)\mu(b) \sum_{q \mid H} \sum_{\substack{(u,v)=1 \\ (u,v) \neq (b,a)}} \mu(u)\mu(v) K_{a,q}(u, v).$$

Thus common content is compressed to divisor-many t, q , but the variable sign survives across the two-dimensional primitive cofactor plane, not along the time variable. Brun–Selberg bounds can thin each prime ray only unsignedly. Explicit squarefree largest-prime rectangles show that the determinant identities admit both signs. The paper proves exact algebra and an exact physical reindexing; it does not prove cancellation in the cofactor plane, a native operator bound, parity breaking, a prime-pair theorem, or a twin-prime result.

Keywords: fixed shift; largest prime factor; residual cofactor; Möbius function; affine prime ray; divisor compression; parity barrier.

Convention. The arithmetic specialization retains one fixed nonzero shift h_0 , every complete native output label, the actual squarefree carrier, and raw atomic normalization. The shift is not set to 2.

MSC 2020: Primary 11N35, 11B25; Secondary 11N36, 15A18.

1 Introduction

The fixed-shift native-operator route separates the missing-high Gram defect into a same-residual part R and an erasure-created part E . TPC-68 identifies the signed operator certificates attached to $R + E$, and TPC-69 proves that the first mixed trace is a cross-key rectangle rather than a one-key correction [8, 9]. At its shortest, the relevant term has two source rows and two complete native keys:

$$\begin{array}{ccc} & m & n \\ \text{erasure key} & \bullet & \bullet \\ \text{same-residual key} & \bullet & \bullet. \end{array}$$

The erasure edge carries the residual sign; the return edge has equal residual labels and contributes no variable target sign after the closed product is formed.

The immediate question is arithmetic rather than matrix-theoretic: *what coordinates does an actual rectangle possess, and along which coordinate can its Möbius sign vary?* The answer is more rigid than a generic two-time correlation. Equal residual labels at the return key compress the two cofactors to a single divisor t of $h_0(n - m)$. At the erasure key, the common cofactor content is another divisor q of the same determinant. After removing this content, the only variable arithmetic sign is $\mu(u)\mu(v)$ on a primitive two-dimensional cofactor plane.

1.1 Main results

Let

$$A_m = mj_R + h_0 = P_m R_m, \quad A_n = nj_R + h_0 = P_n R_n$$

be the two targets at the same-residual key, and let

$$B_m = mj_E + h_0 = Q_m S_m, \quad B_n = nj_E + h_0 = Q_n S_n$$

be the targets at the erasure key. The letters P_m, P_n, Q_m, Q_n denote the actual largest-prime labels of the four physical atoms. The cofactors are R_m, R_n, S_m, S_n .

The paper proves the following exact chain.

- (1) The four targets obey two horizontal determinant equations, two vertical difference equations, and one cross determinant:

$$nA_m - mA_n = nB_m - mB_n = h_0(n - m), \quad A_mB_n - B_mA_n = h_0(n - m)(j_E - j_R).$$

- (2) If $d_m = ga$, $d_n = gb$, $(a, b) = 1$, then the same-residual equality $d_m R_m = d_n R_n$ is equivalent on the squarefree carrier to

$$R_m = bt, \quad R_n = at.$$

The horizontal determinant forces $t \mid h_0(n - m)$. Every fixed- t physical anchor lies on one affine prime ray of spacing abt .

- (3) Writing

$$q = (S_m, S_n), \quad S_m = qu, \quad S_n = qv, \quad (u, v) = 1,$$

the erasure determinant forces $q \mid h_0(n - m)$. The erased residual labels are unequal precisely when

$$(u, v) \neq (b, a).$$

Every fixed- (q, u, v) physical erasure lies on one affine prime ray of spacing quv .

(4) The exact rectangle sign is

$$\mu(a)\mu(b)\mu(u)\mu(v).$$

The common factors g, q disappear, and the sign is constant on each time ray. Therefore an exact grouping of the literal rectangle sum yields a primitive two-dimensional Möbius transform in (u, v) .

The compression is real: t and q run only over divisors of a fixed determinant for a fixed row pair. It is not cancellation. The z -fibers still carry two affine primality conditions, a largest-prime condition, interval restrictions, missing and first-failure masks, complete native labels, and literal complex weights. More importantly, their Möbius sign is frozen. A Brun or Selberg upper-bound sieve can bound the number of prime points on one ray, but it cannot make $\mu(u)\mu(v)$ oscillate along that ray.

1.2 Why this is a useful boundary

The result rules out an attractive but incorrect strategy: reparametrize by time and hope that the target Möbius signs fluctuate along each prime progression. They do not. Any future signed gain must come from one of two sources:

- (i) cancellation across primitive cofactor pairs (u, v) ; or
- (ii) literal complex-phase cancellation already present in the physical kernel.

This locates the next analytic door without claiming that the door is open. A natural continuation is a two-dimensional summation-by-parts criterion for the actual ragged kernel, including its boundary and variation costs.

1.3 Claim discipline

We use the following labels throughout.

- **L0** denotes finite exact arithmetic and algebraic identities.
- **L1** denotes their exact identification with the complete fixed- h_0 native carrier, with no physical label or coefficient removed.
- **L2** denotes a genuinely new cancellation estimate for the literal growing arithmetic family.

This paper proves L0 identities and an L1 reindexing. It proves no L2 estimate. In particular, divisor compression, a ray census, or an unsigned prime upper bound is not fixed- h_0 parity progress.

1.4 Organization

[Section 2](#) fixes the literal two-key carrier. [Section 3](#) proves the four-corner identities. [Section 4](#) compresses the same-residual anchor, and [section 5](#) constructs the primitive erasure coordinates. The ray and sign theorems are proved in [section 6](#). [Section 7](#) gives the exact Möbius-plane transform. [Section 8](#) records the unsigned sieve boundary, and [section 9](#) gives sharp examples. The existing-theorem mismatch, operator ledger, and complete claim boundary occupy the final three sections.

2 The literal two-key interface

We isolate only the information used by the arithmetic reduction. All other physical labels remain attached to the coefficients.

2.1 Rows, times, and four targets

Fix a scale X , one integer $h_0 \neq 0$, and two distinct positive source rows m, n satisfying

$$(mn, h_0) = 1. \tag{1}$$

The TPC carrier also assigns fixed squarefree row divisors

$$d_m \mid m, \quad d_n \mid n.$$

Choose two complete native keys. Their physical times are denoted by j_R and j_E , where R indicates the same-residual key and E the unequal-residual, or erasure, key. Put

$$A_m = mj_R + h_0 = P_m R_m, \quad A_n = nj_R + h_0 = P_n R_n, \quad (2)$$

$$B_m = mj_E + h_0 = Q_m S_m, \quad B_n = nj_E + h_0 = Q_n S_n. \quad (3)$$

Here

$$P_m = P^+(A_m), \quad P_n = P^+(A_n), \quad Q_m = P^+(B_m), \quad Q_n = P^+(B_n)$$

are the literal largest-prime labels selected by the missing-high carrier, and R_m, R_n, S_m, S_n are the corresponding cofactors.

Every active atom considered below lies in the squarefree physical support. Thus all four targets and all four residual labels have nonvanishing Möbius value:

$$\mu(A_m), \mu(A_n), \mu(B_m), \mu(B_n), \mu(d_m R_m), \mu(d_n R_n), \mu(d_m S_m), \mu(d_n S_n) \in \{-1, 1\}. \quad (4)$$

The residual relations defining the two keys are

$$d_m R_m = d_n R_n, \quad (5)$$

$$d_m S_m \neq d_n S_n. \quad (6)$$

Remark 2.1 (Why primitivity is the correct hypothesis). For every integer j ,

$$(m, mj + h_0) = (m, h_0) = 1, \quad (n, nj + h_0) = (n, h_0) = 1.$$

Thus each source row is coprime to every cofactor and largest-prime label on its own side. This elementary fact is what makes the ray spacings in [section 6](#) exact. No coprimality assumption between m and n is required.

2.2 Complete keys and literal weights

A complete native key contains the direct side, external block, opposite row, γ -coordinate, physical time, and every other still orthogonal output label. Equality of complete keys fixes the time, but equality of time alone need not fix the key. We denote the two complete keys by ι_R and ι_E .

At a present key-row corner, let $\beta_{\iota, r} \in \mathbb{C}$ be the literal nonzero coefficient after every physical mask and profile has been imposed. Let $W_r > 0$ be the raw atomic mass of active row r . The residual-sign-free rectangle amplitude is

$$\mathcal{A}(\iota_E, \iota_R; m, n) := \frac{\overline{\beta_{\iota_E, m}} \beta_{\iota_E, n} \overline{\beta_{\iota_R, n}} \beta_{\iota_R, m}}{W_m W_n}. \quad (7)$$

It is defined to be zero if any corner is absent. No sign, phase, or normalization is averaged in this definition.

TPC-69 proves that the oriented rectangle contribution after the fixed source-row gauge is

$$\mu(d_m S_m) \mu(d_n S_n) \mathcal{A}(\iota_E, \iota_R; m, n), \quad (8)$$

and that the full first mixed trace is twice the real part of the sum of these oriented contributions over actual rectangles [9]. We shall reindex (8) exactly. Formula (7), rather than a time-only surrogate, is retained throughout.

2.3 The six-item audit

The specialization used below retains:

- (L1.1) the literal physical coefficients at all four corners;
- (L1.2) the one fixed shift h_0 in all four targets;
- (L1.3) raw atomic row masses W_m, W_n ;
- (L1.4) canonical source rows and complete native output keys;
- (L1.5) the actual active squarefree support, fixed before any extremizer is chosen; and
- (L1.6) zero structural exponent cost for all identities, with every future estimate still subject to the strict 1/400 endpoint ledger.

The reduction does not replace complete keys by times. It only groups already existing keys according to arithmetic coordinates recovered from their atoms.

3 The exact four-corner algebra

Set

$$H := h_0(n - m), \quad \Delta := j_E - j_R. \quad (9)$$

Since $m \neq n$ and $h_0 \neq 0$, one has $H \neq 0$.

Theorem 3.1 (Four-corner identities). *The targets in (2)–(3) satisfy*

$$nA_m - mA_n = H, \quad (10)$$

$$nB_m - mB_n = H, \quad (11)$$

$$B_m - A_m = m\Delta, \quad (12)$$

$$B_n - A_n = n\Delta, \quad (13)$$

$$A_mB_n - B_mA_n = H\Delta. \quad (14)$$

Equivalently, after exposing the four largest primes,

$$nP_mR_m - mP_nR_n = H, \quad (15)$$

$$nQ_mS_m - mQ_nS_n = H, \quad (16)$$

$$P_mR_mQ_nS_n - Q_mS_mP_nR_n = H\Delta. \quad (17)$$

Proof. Substitution gives

$$nA_m - mA_n = n(mj_R + h_0) - m(nj_R + h_0) = h_0(n - m) = H,$$

and the same calculation at j_E proves (11). Subtracting the targets on each row proves (12) and (13). Finally,

$$\begin{aligned} A_mB_n - B_mA_n &= (mj_R + h_0)(nj_E + h_0) - (mj_E + h_0)(nj_R + h_0) \\ &= h_0(n - m)(j_E - j_R) = H\Delta. \end{aligned}$$

The prime-cofactor forms are the same identities after using (2)–(3). □

Corollary 3.2 (A two-time compatibility identity). *Every rectangle satisfies*

$$n(Q_mS_m - P_mR_m) = m(Q_nS_n - P_nR_n). \quad (18)$$

Proof. Subtract (15) from (16). □

Remark 3.3 (Identities are not estimates). The five relations are exact and useful for reindexing. They neither restrict the sign of a rectangle nor provide cancellation. Indeed, [section 9](#) gives two rectangles with identical (m, n, h_0, j_R) and opposite signs, both satisfying all five identities.

4 Same-residual divisor compression

The equal-residual edge is much more rigid than a generic pair of cofactors.

4.1 Primitive row-divisor coordinates

Write

$$g := (d_m, d_n), \quad d_m = ga, \quad d_n = gb. \quad (19)$$

Then $(a, b) = 1$. Since d_m, d_n are squarefree, the three integers g, a, b are squarefree and pairwise coprime.

Theorem 4.1 (Exact anchor compression). *Under (5) and the squarefree support (4), there is a unique positive integer t such that*

$$\boxed{R_m = bt, \quad R_n = at.} \quad (20)$$

Moreover t is squarefree, is coprime to gab , and

$$\boxed{t \mid |H|.} \quad (21)$$

The two largest-prime labels obey the exact line

$$\boxed{nbP_m - maP_n = \frac{H}{t}.} \quad (22)$$

Proof. The residual equality and (19) give

$$aR_m = bR_n.$$

Because $(a, b) = 1$, one has $b \mid R_m$ and $a \mid R_n$. Write

$$R_m = bt_m, \quad R_n = at_n.$$

Substitution gives $t_m = t_n$, proving (20) and uniqueness.

The common residual is

$$d_m R_m = d_n R_n = gabt.$$

It is squarefree by (4). Therefore t is squarefree and coprime to gab . Substituting (20) into (15) gives

$$t(nbP_m - maP_n) = H.$$

This proves (21) and (22). □

Corollary 4.2 (Divisor-many common residual cores). *For fixed m, n, h_0, d_m, d_n , the scale t of an actual same-residual anchor belongs to the finite set*

$$\{t \in \mathbb{N} : t \mid |h_0(n - m)|\}.$$

Hence the number of possible t -values is at most $\tau(|h_0(n - m)|)$. If a physical cofactor window forces $t > |H|$, that row pair has no same-residual anchor in the window.

Remark 4.3 (What is and is not compressed). The parameter t records the entire common residual after the fixed row factors g, a, b are removed. The theorem does not remove the largest-prime constraints, the native-key labels, or the literal coefficient weights. In particular, $\tau(|H|)$ is a count of arithmetic core types, not a bound for the number or total mass of physical rectangles.

4.2 A first cross-determinant consequence

The same-residual substitution also rewrites the cross determinant as

$$tq(bvP_mQ_n - auQ_mP_n) = H\Delta \quad (23)$$

once the erasure coordinates q, u, v of [section 5](#) are introduced. Thus every physical rectangle later satisfies $tq \mid H\Delta$. This compatibility is automatic for actual targets, but it does not imply $tq \mid H$ and it does not determine a Möbius sign.

5 Primitive coordinates at the erasure key

The unequal-residual edge has one common cofactor content and two primitive residual directions.

Definition 5.1 (Primitive erasure coordinates). Put

$$q := (S_m, S_n), \quad S_m = qu, \quad S_n = qv. \quad (24)$$

Then $(u, v) = 1$.

Theorem 5.2 (Erasure divisor and prime line). *On the squarefree physical support:*

- (i) q, u, v are squarefree and pairwise coprime;
- (ii) q is coprime to gab , u is coprime to ga , and v is coprime to gb ;
- (iii)

$$\boxed{q \mid |H|} \quad (25)$$

- (iv) the largest-prime labels obey

$$\boxed{nuQ_m - mvQ_n = \frac{H}{q}} \quad (26)$$

- (v) the unequal-residual condition is exactly

$$\boxed{(u, v) \neq (b, a)}. \quad (27)$$

Proof. The residual labels are

$$d_m S_m = gaqu, \quad d_n S_n = gbqv.$$

Both are squarefree by (4). Since $(u, v) = 1$ by construction, the stated squarefreeness and coprimality conditions follow.

Substitute (24) into (16):

$$q(nuQ_m - mvQ_n) = H.$$

This proves (25) and (26).

Finally,

$$d_m S_m = d_n S_n \iff au = bv.$$

If $au = bv$, then $(a, b) = 1$ gives $a \mid v$ and $b \mid u$. Write $u = bk$, $v = al$. The equality gives $k = \ell$, while $(u, v) = 1$ forces $k = \ell = 1$. Thus equality holds precisely at $(u, v) = (b, a)$, proving (27). $\square$

Corollary 5.3 (Two divisor coordinates, one primitive plane). *For a fixed row pair, both common-content variables satisfy*

$$t, q \mid |h_0(n - m)|.$$

The anchor variable t determines its two cofactors completely through (bt, at) . By contrast, after q is fixed, the erasure edge still ranges over primitive pairs (u, v) , except for the single forbidden pair (b, a) .

Proposition 5.4 (Primitive four-corner equations). *After (20) and (24), the vertical and cross relations become*

$$quQ_m - btP_m = m\Delta, \quad (28)$$

$$qvQ_n - atP_n = n\Delta, \quad (29)$$

$$tq(bvP_mQ_n - auQ_mP_n) = H\Delta. \quad (30)$$

Proof. The first two formulas are (12)–(13) after substitution. The third is (17) in the same coordinates. $\square$

Remark 5.5 (The cross equation does not collapse the plane). The variables $P_m, P_n, Q_m, Q_n, \Delta$ change with the physical keys. Consequently (30) is a compatibility condition on each point, not a fixed conic that removes the two-dimensional (u, v) -sum. Any such further reduction would require additional physical information.

6 Affine rays and frozen arithmetic signs

The divisor coordinates determine exact congruence classes for the two physical times. The following statements concern physical solutions: primality, largest-prime, interval, native-key, and mask conditions may delete points from the displayed affine rays, but cannot create points off them.

6.1 The anchor ray

Theorem 6.1 (Fixed- t anchor ray). *Fix m, n, h_0, d_m, d_n and a divisor t supplied by theorem 4.1. If one physical anchor solution $(j_{R,0}, P_{m,0}, P_{n,0})$ exists, then every other physical anchor solution with the same t is uniquely of the form*

$$j_R = j_{R,0} + abt z, \quad (31)$$

$$P_m = P_{m,0} + ma z, \quad (32)$$

$$P_n = P_{n,0} + nb z \quad (33)$$

for an integer z .

Proof. Take two fixed- t anchor solutions and write differences with a δ . From

$$mj_R + h_0 = btP_m$$

one obtains

$$m\delta j_R = bt\delta P_m.$$

By remark 2.1, $(m, bt) = 1$, so $bt \mid \delta j_R$. Likewise $at \mid \delta j_R$. Since a, b, t are pairwise coprime,

$$\text{lcm}(at, bt) = abt.$$

Thus $\delta j_R = abt z$. Substitution gives $\delta P_m = ma z$ and $\delta P_n = nb z$. Uniqueness is immediate. $\square$

Corollary 6.2 (Anchor census). *If the allowed anchor-time interval has length J_R , then for fixed t*

$$\#\{j_R \text{ in the physical carrier with this } t\} \leq 1 + \frac{J_R}{abt}. \quad (34)$$

6.2 The erasure ray

Theorem 6.3 (Fixed- (q, u, v) erasure ray). *Fix m, n, h_0 and a primitive erasure triple (q, u, v) from [theorem 5.2](#). If one physical erasure solution $(j_{E,0}, Q_{m,0}, Q_{n,0})$ exists, then every other physical erasure solution with the same (q, u, v) is uniquely of the form*

$$j_E = j_{E,0} + quv z, \quad (35)$$

$$Q_m = Q_{m,0} + mv z, \quad (36)$$

$$Q_n = Q_{n,0} + nu z \quad (37)$$

for an integer z .

Proof. For two solutions,

$$m \delta j_E = qu \delta Q_m, \quad n \delta j_E = qv \delta Q_n.$$

By [remark 2.1](#), $(m, qu) = (n, qv) = 1$. Hence $qu \mid \delta j_E$ and $qv \mid \delta j_E$. The integers q, u, v are pairwise coprime, so their least common multiple is quv . Writing $\delta j_E = quv z$ gives the two prime increments. $\square$

Corollary 6.4 (Erasure census). *If the allowed erasure-time interval has length J_E , then*

$$\#\{j_E \text{ in the physical carrier with fixed } q, u, v\} \leq 1 + \frac{J_E}{quv}. \quad (38)$$

6.3 The sign theorem

Theorem 6.5 (Primitive cofactor sign). *The residual sign in the literal rectangle contribution [\(8\)](#) is*

$$\boxed{\mu(d_m S_m) \mu(d_n S_n) = \mu(a) \mu(b) \mu(u) \mu(v)}. \quad (39)$$

Equivalently, the product of the four target Möbius signs is

$$\boxed{\mu(A_m) \mu(A_n) \mu(B_m) \mu(B_n) = \mu(a) \mu(b) \mu(u) \mu(v)}. \quad (40)$$

Proof. Using the squarefree and coprimality conclusions of [theorem 5.2](#),

$$\begin{aligned} \mu(d_m S_m) \mu(d_n S_n) &= \mu(ga qu) \mu(gb qv) \\ &= \mu(g)^2 \mu(q)^2 \mu(a) \mu(b) \mu(u) \mu(v), \end{aligned}$$

which proves [\(39\)](#).

For the target signs, each largest prime is coprime to its cofactor and contributes one minus sign:

$$\mu(A_m) \mu(A_n) \mu(B_m) \mu(B_n) = \mu(R_m) \mu(R_n) \mu(S_m) \mu(S_n).$$

The anchor factors give

$$\mu(bt) \mu(at) = \mu(a) \mu(b),$$

and the erasure factors give

$$\mu(qu) \mu(qv) = \mu(u) \mu(v).$$

This proves [\(40\)](#). $\square$

Corollary 6.6 (Frozen sign on every prime ray). *For fixed rows m, n and fixed primitive erasure coordinates (q, u, v) , the Möbius sign of every physical point on [\(35\)–\(37\)](#) is the same. There is no arithmetic sign oscillation in the ray variable z .*

Remark 6.7 (What can still oscillate). The literal amplitude $\mathcal{A}$ can vary with the complete native key and with z . Its complex phases may cancel. The theorem says only that such cancellation cannot be attributed to a varying target Möbius sign inside one fixed (q, u, v) -ray.

7 The exact primitive cofactor-plane transform

We now group the literal rectangle sum without dropping any native label.

7.1 Anchored rectangles

Fix $m < n$. A *same-residual anchor* $\mathfrak{a}$ consists of:

$$\mathfrak{a} = (m, n, \iota_R, j_R, P_m, P_n, R_m, R_n, t)$$

together with its two literal corner coefficients. Every entry is recovered from an actual complete native key satisfying (5). Let $\mathcal{R}_{m,n}$ be the finite set of such anchors.

For a fixed anchor, let $\mathcal{E}(\mathfrak{a})$ be the finite set of actual erasure keys ι_E completing the other two corners and satisfying (6). Define the oriented anchored sum

$$\mathfrak{X}_{\mathfrak{a}} := \sum_{\iota_E \in \mathcal{E}(\mathfrak{a})} \mu(d_m S_m(\iota_E)) \mu(d_n S_n(\iota_E)) \mathcal{A}(\iota_E, \iota_R; m, n). \quad (41)$$

With the oriented convention of section 2,

$$\text{tr}(ER) = 2 \text{Re} \sum_{m < n} \sum_{\mathfrak{a} \in \mathcal{R}_{m,n}} \mathfrak{X}_{\mathfrak{a}}. \quad (42)$$

7.2 The literal ray kernel

Fix $\mathfrak{a}$, a positive divisor $q \mid |H|$, and positive coprime integers u, v . If an erasure ray with these coordinates is nonempty, choose one arithmetic representative

$$(j_{E,0}, Q_{m,0}, Q_{n,0}).$$

Every actual key in that fiber has the form

$$j_E = j_{E,0} + quvz, \quad Q_m = Q_{m,0} + mvz, \quad Q_n = Q_{n,0} + nuvz.$$

Define

$$K_{\mathfrak{a},q}(u, v) := \sum_{z \in \mathbb{Z}} \sum_{\substack{\iota_E \in \mathcal{E}(\mathfrak{a}) \\ j(\iota_E) = j_{E,0} + quvz \\ (S_m(\iota_E), S_n(\iota_E)) = (qu, qv)}} \mathcal{A}(\iota_E, \iota_R; m, n). \quad (43)$$

If no representative exists, set $K_{\mathfrak{a},q}(u, v) = 0$. This definition is independent of the chosen representative, since changing it only translates z .

The kernel in (43) retains:

- both affine primality and largest-prime conditions;
- all squarefree and coprimality conditions not displayed outside the kernel;
- the actual finite cofactor and time ranges;
- side, block, opposite-row, γ , missing, first-failure, and endpoint masks;
- all four literal complex coefficients; and
- raw atomic normalization by $W_m W_n$.

It is therefore a ragged, generally complex kernel. No positivity, smoothness, product structure, or low variation is assumed.

Theorem 7.1 (Exact primitive-plane transform). *For every same-residual anchor $\mathfrak{a}$,*

$$\boxed{\mathfrak{X}_{\mathfrak{a}} = \mu(a)\mu(b) \sum_{q \mid |H|} \sum_{\substack{u, v \geq 1 \\ (u, v) = 1 \\ (u, v) \neq (b, a)}} \mu(u)\mu(v) K_{\mathfrak{a},q}(u, v).} \quad (44)$$

All sums are finite because the physical carrier is finite.

Proof. Every $\iota_E \in \mathcal{E}(\mathbf{a})$ has a unique triple

$$q = (S_m, S_n), \quad u = S_m/q, \quad v = S_n/q.$$

By [theorem 5.2](#), this triple satisfies

$$q \mid |H|, \quad (u, v) = 1, \quad (u, v) \neq (b, a).$$

Conversely, the kernel in (43) contains exactly the actual erasure keys with the displayed coordinates. Thus the right-hand side partitions $\mathcal{E}(\mathbf{a})$ without omission or multiplicity. Finally, [theorem 6.5](#) replaces the residual sign on each part by $\mu(a)\mu(b)\mu(u)\mu(v)$. $\square$

Corollary 7.2 (Complete rectangle transform). *The first mixed trace has the exact expansion*

$$\boxed{\operatorname{tr}(ER) = 2 \operatorname{Re} \sum_{m < n} \sum_{\mathbf{a} \in \mathcal{R}_{m,n}} \mu(a)\mu(b) \sum_{q \mid |h_0(n-m)|} \sum_{\substack{(u,v)=1 \\ (u,v) \neq (b,a)}} \mu(u)\mu(v) K_{\mathbf{a},q}(u, v).} \quad (45)$$

7.3 The exact location of possible cancellation

For a fixed primitive pair, [corollary 6.6](#) gives

$$|\mu(a)\mu(b)\mu(u)\mu(v)K_{\mathbf{a},q}(u, v)| = |K_{\mathbf{a},q}(u, v)|. \quad (46)$$

Thus grouping or estimating the z -fiber cannot by itself create Möbius cancellation. It may create complex-phase cancellation inside K , but the arithmetic sign is a scalar outside the fiber.

A genuine signed estimate must control a sum of the form

$$\sum_{\substack{u \sim U, v \sim V \\ (u,v)=1 \\ (u,v) \neq (b,a)}} \mu(u)\mu(v)K_{\mathbf{a},q}(u, v) \quad (47)$$

uniformly over the actual growing family of rows, anchors, divisors, blocks, and coefficient profiles. Small or boundary values of u, v are part of the physical support; the notation $u \sim U, v \sim V$ is only a possible later dyadic partition, not a hypothesis of this paper.

Remark 7.3 (Zero extension and boundary cost). One may extend $K_{\mathbf{a},q}$ by zero to a rectangle in $\mathbb{Z}_{\geq 1}^2$ and absorb the primitive and excluded-point indicators into the kernel. This is algebraically valid. Analytically, every jump created by the ragged support, hard prime-line conditions, or a missing mask contributes to the two-dimensional boundary or variation. Such costs cannot be discarded in a later summation-by-parts argument.

7.4 A useful but unproved L2 target

Let

$$\|K\|_{\ell^1(U,V)} := \sum_{\substack{u \sim U \\ v \sim V}} |K(u, v)|.$$

A representative L2 input would be a number $\eta_X = o(1)$ such that

$$\left| \sum_{\substack{u \sim U, v \sim V \\ (u,v)=1 \\ (u,v) \neq (b,a)}} \mu(u)\mu(v)K_{\mathbf{a},q}(u, v) \right| \leq \eta_X \|K_{\mathbf{a},q}\|_{\ell^1(U,V)} \quad (48)$$

in the required uniform physical scope. No estimate of the form (48) is proved here. It is displayed to make the next arithmetic door falsifiable.

8 Unsigned ray bounds and the sieve boundary

The affine rays expose two prime forms, so classical upper-bound sieve technology can count candidate points. This is useful support control, but it is sign-blind.

8.1 Nonproportional prime forms

On a fixed anchor ray define

$$L_{m,R}(z) = P_{m,0} + ma z, \quad L_{n,R}(z) = P_{n,0} + nb z.$$

Equation (22) gives

$$nbL_{m,R}(z) - maL_{n,R}(z) = H/t \neq 0. \quad (49)$$

Hence the forms are nonproportional. They are primitive because $(P_{m,0}, ma) = (P_{n,0}, nb) = 1$, a consequence of (1).

Likewise, on a fixed erasure ray,

$$L_{m,E}(z) = Q_{m,0} + mv z, \quad L_{n,E}(z) = Q_{n,0} + nu z,$$

and

$$nuL_{m,E}(z) - mvL_{n,E}(z) = H/q \neq 0. \quad (50)$$

These two forms need not be primitive a priori because, for example, $Q_{m,0}$ can divide the cross-cofactor v . If one form is imprimitive, however, it assumes a positive prime value at most once. Consequently every erasure ray containing two distinct simultaneous prime points has two primitive forms; singleton rays are already covered by the trivial census.

Proposition 8.1 (Fixed-ray Brun–Selberg boundary). *Fix either pair of primitive, nonproportional affine forms above. If the pair is admissible, the upper-bound sieve gives, for an interval $\mathcal{I} \subset \mathbb{Z}$ of length Z ,*

$$\#\{z \in \mathcal{I} : L_1(z), L_2(z) \text{ are prime}\} \ll_{L_1, L_2} 1 + \frac{Z}{(\log(2+Z))^2}. \quad (51)$$

If the pair has a fixed local obstruction, simultaneous prime values are confined to $O_{L_1, L_2}(1)$ exceptional points.

Proof. This is the standard dimension-two upper-bound sieve for a product of two primitive, nonproportional affine forms; see, for example, the Selberg sieve treatment in [3]. A fixed local obstruction forces at least one form to equal one of finitely many local primes at every simultaneous prime point. $\square$

Remark 8.2 (Uniformity is not supplied). The constant in (51) is allowed to depend on the ray data. The physical coefficients m, n, a, b, u, v grow with X , the ray length can be as short as

$$Z \asymp \frac{J_R}{abt} \quad \text{or} \quad Z \asymp \frac{J_E}{quv},$$

and the carrier contains additional largest-prime and hard-mask conditions. A uniform weighted bound over the complete family is a separate theorem.

8.2 Why an upper-bound sieve does not move the sign

Even a perfectly uniform version of (51) would estimate only the absolute population of one ray. By corollary 6.6,

$$\begin{aligned} \sum_z \mu(a)\mu(b)\mu(u)\mu(v) \mathcal{A}_z \\ = \mu(a)\mu(b)\mu(u)\mu(v) \sum_z \mathcal{A}_z. \end{aligned} \quad (52)$$

If the amplitudes $\mathcal{A}_z$ are nonnegative, taking an upper-bound sieve loses no arithmetic cancellation because none exists inside the fiber. For complex amplitudes, any gain in $\sum_z \mathcal{A}_z$ is a phase theorem, not a Möbius theorem.

Corollary 8.3 (Unsigned support gain is not parity progress). *The divisor counts, ray spacings, and Brun–Selberg upper bounds in this paper can reduce an unsigned majorant. None of them proves cancellation in (47); hence none is a fixed-shift parity-breaking estimate.*

9 Sharp examples and no-go models

The following examples separate exact determinant geometry from signed cancellation.

9.1 One anchor, both signs

Example 9.1 (Opposite signs with the same rows and anchor). *Take*

$$m = 3, \quad n = 5, \quad h_0 = 1, \quad d_m = d_n = 1.$$

Then $g = a = b = 1$ and $H = 2$. Choose $j_R = 2$. The anchor targets are

$$A_m = 7 = 7 \cdot 1, \quad A_n = 11 = 11 \cdot 1.$$

Thus $R_m = R_n = 1$, the residuals agree, and $t = 1$.

First take $j_E = 4$. Then

$$B_m = 13 = 13 \cdot 1, \quad B_n = 21 = 7 \cdot 3.$$

Hence

$$q = 1, \quad (u, v) = (1, 3),$$

and the rectangle sign is

$$\mu(1)\mu(1)\mu(1)\mu(3) = -1.$$

The cross determinant is

$$7 \cdot 21 - 13 \cdot 11 = 4 = H(j_E - j_R).$$

Now take $j_E = 18$. Then

$$B_m = 55 = 11 \cdot 5, \quad B_n = 91 = 13 \cdot 7,$$

where the displayed primes are the largest prime factors. Thus

$$q = 1, \quad (u, v) = (5, 7),$$

and the sign is

$$\mu(5)\mu(7) = +1.$$

Again all four targets are squarefree and

$$7 \cdot 91 - 55 \cdot 11 = 32 = H(j_E - j_R).$$

Therefore the four-corner identities and the divisor conditions admit both signs even with the rows and same-residual anchor fixed.

Remark 9.2 (Scope of the example). The example realizes the exact squarefree largest-prime arithmetic dictionary. It is not asserted to satisfy every scale-dependent missing-high, block, or endpoint mask of the full TPC carrier. Its role is sharp: no argument using only the identities proved in [section 3](#) can determine the sign.

9.2 Two points on one frozen ray

Example 9.3 (A nontrivial ray with constant sign). *Keep the same rows and anchor, and the erasure coordinates*

$$(q, u, v) = (1, 1, 3).$$

The point $j_E = 4$ above has

$$(Q_m, Q_n) = (13, 7).$$

Theorem 6.3 gives the candidate ray

$$j_E = 4 + 3z, \quad Q_m = 13 + 9z, \quad Q_n = 7 + 5z.$$

At $z = 2$, this gives

$$j_E = 10, \quad Q_m = 31, \quad Q_n = 17,$$

and indeed

$$3 \cdot 10 + 1 = 31 \cdot 1, \quad 5 \cdot 10 + 1 = 17 \cdot 3.$$

Both $z = 0$ and $z = 2$ are squarefree largest-prime points, and both have sign -1 . The intermediate candidate $z = 1$ is deleted because $Q_m = 22$ is not prime. Thus primality thins the ray but does not change its arithmetic sign.

9.3 Kernel-level obstruction

Proposition 9.4 (No cancellation from the coordinate change alone). *For every finite set*

$$\Omega \subset \{(u, v) \in \mathbb{N}^2 : (u, v) = 1, (u, v) \neq (b, a)\}$$

on which $\mu(u)\mu(v) = \sigma \in \{-1, 1\}$, there is a nonnegative kernel K supported on Ω for which

$$\left| \sum_{(u,v)} \mu(u)\mu(v)K(u,v) \right| = \sum_{(u,v)} K(u,v). \quad (53)$$

Proof. Choose any nonzero nonnegative function K supported on Ω . The signed sum is $\sigma \sum K$, proving the identity. $\square$

The proposition is an abstract sharpness model, not a claim about the actual physical kernel. It proves that divisor compression and primitive coordinates alone cannot imply a signed saving. A future theorem must use additional regularity, distribution, or phase structure of the literal kernel.

9.4 Three common invalid inferences

- (1) From $t, q \mid H$ one cannot infer that there are only divisor-many rectangles: the primitive pair (u, v) , the ray points, and the complete native labels remain.
- (2) From $t \mid H$ and $q \mid H$ one cannot infer $tq \mid H$. The exact cross relation only gives $tq \mid H\Delta$.
- (3) From two nonproportional prime forms one cannot infer signed cancellation. Upper-bound sieves control their population, while the sign $\mu(u)\mu(v)$ is constant on the whole ray.

10 Existing-theorem boundary

The primitive-plane transform changes the analytic shape of the problem. It does not solve it. We record which standard inputs are genuinely available and which interfaces remain unmatched.

10.1 Unconditional Mertens cancellation is available only through the kernel

The prime number theorem is equivalent to

$$M(x) := \sum_{n \leq x} \mu(n) = o(x). \quad (54)$$

Consequently a sufficiently regular product-like kernel can be treated by one- or two-dimensional summation by parts. Formula (44) makes this possibility visible: the variable sign is the separable product $\mu(u)\mu(v)$.

However, (54) does not control

$$\sum_{u,v} \mu(u)\mu(v)K(u,v)$$

for an arbitrary bounded or sign-correlated kernel. The literal $K_{a,q}$ includes prime-ray solvability, largest-prime conditions, ragged cofactor ranges, missing masks, and complex weights. A proof must charge the boundary and variation produced by these features. This is the precise reason the Mertens theorem is not silently inserted into [theorem 7.1](#).

10.2 Nearby correlation theorems

TPC-33 gives a detailed comparison between the physical fixed-determinant families and existing Chowla–Elliott technology [6]. The relevant theorem forms include:

- Tao’s logarithmically averaged two-point theorem for fixed nonproportional affine forms [5];
- the Matomäki–Radziwiłł–Tao averaged Chowla theorem, which includes an additional averaging parameter [4]; and
- the Frantzikinakis–Host mean-zero theorem for fixed independent linear forms on a full multidimensional box [1].

These results are strong precedents, but none is invoked here as a bound for (47). As stated in the cited sources, they do not simultaneously supply all of:

- (i) the actual finite, ragged (u,v) -support;
- (ii) coefficients and ray slopes growing with the physical scale;
- (iii) the hard two-prime and largest-prime masks inside K ;
- (iv) every complete native block and literal complex weight; and
- (v) the uniform row, anchor, and divisor scope needed by the operator ledger.

This statement compares the displayed theorem forms with the present interface; it is not a claim that no future combination or refinement of those methods can apply.

10.3 The classical parity boundary

Brun–Selberg sieves provide the unsigned ray estimate in [proposition 8.1](#). They do not distinguish the two values of $\mu(u)\mu(v)$. This is the familiar sign-sensitive boundary of sieve theory; see the classical parity discussion in [2]. Our coordinate change does not evade that boundary. It identifies the exact two-dimensional location at which an additional input would have to act.

Table 1: Available inputs and the remaining primitive-plane interface.

Input	What it controls	What remains here
Prime number theorem	$M(x) = o(x)$	two-dimensional boundary and variation of the actual ragged kernel
Brun–Selberg upper bound	population on one fixed prime ray	sign is frozen on that ray; growing-family weighted uniformity
Two-point logarithmic Chowla	fixed affine forms under logarithmic averaging	physical dyadic kernel, growing slopes, native masks, and operator scope
Full-box multiplicative averages	fixed independent forms on a full box	determinant-coupled prime-ray kernel on a nonrectangular support

10.4 A precise positive route

The literature boundary is not purely negative. It suggests a concrete test. Zero-extend $K_{\mathfrak{a},q}$ on each dyadic cofactor box and prove a two-dimensional Abel identity whose right-hand side contains:

- (1) products of Mertens partial sums;
- (2) mixed finite differences of $K_{\mathfrak{a},q}$;
- (3) four edge terms and four corner terms; and
- (4) explicit jumps created by primitive, prime-line, and missing indicators.

If the resulting variation and boundary total is smaller than the literal ℓ^1 mass in the required uniform scope, it yields a genuine L2 saving. If the hard masks force variation comparable to the full mass, this route stops cleanly and the failure is itself a rigorous boundary result.

11 From one primitive plane back to the operator

The exact transform concerns the first mixed trace. We state the remaining interfaces before any arithmetic gain can influence the fixed-shift endpoint.

11.1 The hierarchy of scopes

For a possible estimate of (44), distinguish:

- (i) one dyadic (U, V) -box for one fixed $(m, n, \mathfrak{a}, q)$;
- (ii) all cofactor boxes for that anchor;
- (iii) all anchors and erasure keys for one row pair;
- (iv) all partners incident to one active row;
- (v) the complete trace sum; and
- (vi) a full local-star or operator-norm estimate.

These scopes are not interchangeable. A small complete trace can result from cancellation between rows while one local star remains large. A fixed-anchor saving can be lost when the number or total mass of anchors grows.

11.2 The rectangle is only the linear erasure sector

The first mixed moment is

$$2 \operatorname{tr}(ER).$$

Higher even moments contain one-erasure sectors

$$2q \operatorname{tr}(ER^{2q-1})$$

and sectors with two or more erasure edges. TPC-69 proves that the full two-erasure sector in every even moment is nonnegative [9]. Therefore a saving for the rectangle sum does not bound a complete even moment, and the quadratic sector cannot be used as a negative correction.

Even a theorem of the form

$$\text{tr}(ER) = o\left(\sum_{\text{rectangles}} |\mathcal{A}|\right)$$

would close only the first signed scalar door. To reach a native operator estimate, one must either:

- extend primitive-plane control to the local stars and all relevant mixed words; or
- find a different certificate that converts the same literal arithmetic input to $\|R + E\|$.

11.3 Mass and shorting gates

Let M_X be the missing-native map and D_M its raw atomic row mass. The normalized missing Gram has the form

$$D_M^{\dagger/2} M_X^* M_X D_M^{\dagger/2} = I + R + E$$

on the active row space [7, 8]. A hypothetical operator estimate for $R + E$ still does not finish reassembly. It must be combined with:

- (1) a missing-mass comparison in the literal coefficient norm;
- (2) a lower-frame estimate for the represented map; and
- (3) a negligible nuisance shorting penalty.

In the notation of the existing route, the shorted Gram is

$$G_{\text{sh}} = G_{\text{dir}} - C_N, \quad C_N \succeq 0,$$

and $\|C_N\| = o(1)$ is a separate physical statement. The primitive cofactor plane neither identifies C_N nor bounds it.

11.4 Endpoint exponent ledger

All identities in this paper cost zero fixed power of X . Future arguments may introduce losses for:

$$\lambda_{\text{plane}}, \quad \lambda_{\text{anchor}}, \quad \lambda_{\text{native}}, \quad \lambda_{\text{mass}}, \quad \lambda_{\text{rep}}, \quad \lambda_{\text{nuis}}.$$

Only losses proved in a common literal physical scope may be added. The total must remain strictly below

$$\frac{1}{400}.$$

Equality is insufficient for the present endpoint. A valid theorem outside this budget may still be valuable, but it cannot be promoted to the fixed-shift endpoint implication.

11.5 No fixed-shift arithmetic promotion

The paper does not establish that the primitive-plane transform is smaller than its absolute majorant. Consequently it supplies no new exponent for the literal physical coefficients, no bound for $\|R + E\|$, and no improvement in the endpoint ledger. Its forward value is architectural: it closes an exact reindexing interface and permanently rules out time-ray Möbius oscillation as the missing mechanism.

12 Claims, stops, and the next gate

12.1 L0 results proved here

For four squarefree largest-prime targets attached to two primitive rows:

(1) the five four-corner identities (10)–(14);

(2) the same-residual normal form

$$R_m = bt, \quad R_n = at, \quad t \mid |h_0(n - m)|;$$

(3) the primitive erasure normal form

$$S_m = qu, \quad S_n = qv, \quad q \mid |h_0(n - m)|;$$

(4) the exact erasure criterion

$$d_m S_m \neq d_n S_n \iff (u, v) \neq (b, a);$$

(5) the anchor and erasure affine-ray parametrizations and their exact census bounds;

(6) the sign identity

$$\mu(d_m S_m) \mu(d_n S_n) = \mu(a) \mu(b) \mu(u) \mu(v);$$

(7) the fact that this sign is frozen along every fixed (q, u, v) -ray; and

(8) the exact finite primitive-plane partition of an anchored rectangle sum.

12.2 L1 identification

The L0 coordinates are identified with the literal TPC two-key rectangle by retaining:

- the fixed nonzero h_0 in every target;
- the actual largest prime and cofactor at all four corners;
- every complete native key and actual support condition;
- all four literal complex coefficients;
- raw atomic row normalization; and
- the full physical kernel $K_{a,q}(u, v)$, with zero coefficient at every absent corner.

Consequently (45) is an exact L1 reindexing of $\text{tr}(ER)$, not a model average.

12.3 L2 statements not proved

This paper does not prove:

- cancellation in one primitive cofactor box;
- a uniform two-dimensional Mertens or variation bound for the actual kernel;
- a saving after summing all q, t , anchors, keys, or row pairs;
- a local-star bound or a complete mixed-moment bound;
- a subpower or vanishing bound for $\|R + E\|$;
- missing-mass, represented-frame, or nuisance-shortening estimates;
- any fixed- h_0 parity-breaking estimate;
- a prime-pair theorem, a twin-prime theorem, or a positive lower bound for prime pairs.

12.4 Kill criteria

- (K1) **Nonprimitive rows.** The exact ray spacings use $(mn, h_0) = 1$. Applying them outside that scope without extracting the additional gcd changes the parametrization.
- (K2) **Nonsquarefree support.** The sign formula is a $\{-1, 1\}$ -identity on the active squarefree carrier. Terms with vanishing Möbius factors must be removed or treated separately.
- (K3) **Time-only compression.** Replacing complete native keys by their times deletes orthogonal labels and estimates a different operator.
- (K4) **False ray randomness.** The target Möbius sign is constant on a fixed (q, u, v) -ray. Random-sign or time-oscillation heuristics cannot certify a saving there.
- (K5) **Divisor-count overclaim.** Although $t, q \mid H$, the primitive pair, ray points, and complete keys remain. The number of rectangles is not bounded by $\tau(H)^2$.
- (K6) **Silent rectangularization.** Zero-extending the ragged support is allowed only if every new boundary and variation term is retained.
- (K7) **Unsigned-to-signed promotion.** A Brun–Selberg upper bound is an absolute population estimate, not cancellation of $\mu(u)\mu(v)$.
- (K8) **Trace-to-operator promotion.** A small $\text{tr}(ER)$ does not imply a local-star, moment, or operator-norm bound.
- (K9) **Reassembly mismatch.** A missing-native estimate does not prove the mass, represented-frame, or nuisance shorting gates.
- (K10) **Endpoint stop.** Any combined literal loss at least $1/400$ stops the present endpoint implication.

12.5 The next specified gate

The next falsifiable question is not another coordinate change. It is:

For the zero-extended literal kernels $K_{a,q}$, can two-dimensional summation by parts combine unconditional Mertens cancellation with a uniformly subcritical boundary-and-variation total?

A complete next step should:

- (i) derive an exact discrete two-dimensional Abel identity;
- (ii) separate interior mixed variation from four edges and four corners;
- (iii) charge the primitive condition, the excluded point, ragged cofactor support, prime-line masks, and missing masks literally;
- (iv) compare the resulting certificate with $\sum |K_{a,q}|$ in every dyadic box; and
- (v) either prove a uniform saving or give an actual-kernel lower bound showing that the variation route cannot save.

If the certificate succeeds only for smoothed, averaged, or frozen kernels, that result must remain in its weaker scope. If it fails because hard physical masks have full variation, the failure is a publishable negative boundary and the route should return to literal complex phases or direct shorted reassembly.

12.6 Status line

TPC–70 compresses both common-content variables to divisors of the fixed row-pair determinant and identifies the surviving sign as a primitive two-dimensional Möbius product. It also proves that time-ray sign oscillation is impossible. This is a genuine L0/L1 advance and a sharper analytic target. It is not L2 cancellation and gives no evidence that the parity barrier has been crossed.

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